

Exact Post-Transient Coarse Factorization in a Reduced Male *Drosophila* CNS Model

Reached-Set Restriction, Safe Forgetting, and Internal Transition-Layer Contrast

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This paper is Paper XVII of the RIME program, an independently scoped computational follow-up to Paper XVI. It does not extend the SOF protocol line and does not reopen Paper XVI's published claim surface.

Abstract

Problem. A lossy observation may fail to determine even one observed successor on an initial microscopic domain. It may nevertheless support an exact one-step factorization on states reached later by the dynamics. Memory restoration and reached-set restriction must be distinguished on a common source-time support.

Approach. We study the exact rational dynamics and superclass-mean observation used in Paper XVI. A declared 711-source domain spans all 25 nonempty source strata, with a four-step horizon. Exact audits compare history orders on the common support $t \in \{2, 3\}$, decompose its non-singleton fibers, and test the terminal image. A separate pair census records raw signed-drive, clipping, and coarse-residual differences under zero input and noise.

Results. The apparent order-two closure in the order-specific output is a left-boundary effect. On the common support, orders zero, one, and two induce the same noninjective, successor-compatible partition of 1,422 windows into 1,228 values. All 51 sources in the six unsafe $t = 1$ fibers become coarsely singleton at $t = 2$; five disjoint cohorts of sizes 51, 25, 19, 4, 3 remain non-singleton and successor-compatible. The terminal $t = 4$ image lies outside the verified common coarse image, so the certificate gives one internal shift, not an endomap. In 6,975 exact pair transitions, two clipping-sensitive predicates separate all 570 transient-obstruction pairs from all 1,755 persistent-safe pairs at both $2 \rightarrow 3$ and $3 \rightarrow 4$. No single registered predicate separates the classes at the defining $1 \rightarrow 2$ transition. An exact cache audit covers every member of all ten non-singleton fibers: their clipped network drive is zero, so common linear leakage suffices to preserve their one-step coarse equality.

Boundary. The result is an exact finite Computational Certificate on a reduced domain and horizon $T = 4$. It does not establish a coarse endomap, minimal memory, full-domain closure, a causal trigger for the initial split, or physiological brain-state dynamics. Validation is local exact replay, not independent replication. The noninjective factorization does not demonstrate active compensation or nonzero-drive dynamic compression.

Keywords. connectome; coarse-graining; coarse factorization; reached set; quotient dynamics; exact rational computation; transient separation; clipping; MaleCNS

Notation Table

symbol	meaning
X	microscopic state space of the registered deterministic model
$F_Y : X \rightarrow X$	registered exact microscopic update induced by the signed carrier Y
D_0	declared reduced set of 711 unit-state sources
$x_{a,t} = F_Y^t(e_a)$	microscopic state from source $a \in D_0$ at time t
$O : X \rightarrow Z$	registered superclass-mean observation
$z_{a,t} = O(x_{a,t})$	exact coarse observation
$H_h(a, t)$	newest-first history $(z_{a,t}, \dots, z_{a,t-h})$
$S(a, t)$	exact successor observation $z_{a,t+1}$
Ω_h	original order-specific source-time window domain
Ω^{com}	common support $D_0 \times \{2, 3\}$
Π_h	partition of the selected source-time support into equal- H_h fibers
N_h	number of registered windows
K_h	number of distinct exact history values
M_h	largest exact history-fiber cardinality
$R_{2,3}$	registered post-transient reached-set slices at $t = 2, 3$
$\mathcal{Z}_{2,3} = O(R_{2,3})$	verified common coarse image
$u_{a,t} = Y^\top x_{a,t}$	exact raw signed drive before clipping
$C(u)$	coordinatewise clipping $\min\{1, \max\{0, u\}\}$
r_{raw}	coarse residual $O(u^a) - O(u^b)$ for an oriented source pair
r_{clip}	coarse residual $O(C(u^a)) - O(C(u^b))$ after clipping
r_{corr}	exact clipping correction $r_{\text{clip}} - r_{\text{raw}}$

1 Introduction

A coarse observation does not automatically support a closed dynamical state. Given microscopic dynamics $F : X \rightarrow X$ and an observation $O : X \rightarrow Z$, a one-step coarse factorization exists on a declared domain only when $O(Fx)$ is constant on every fiber of O . Paper XVI applied this elementary criterion to a signed MaleCNS connectome model and found exact first-order and one-lag obstructions on its registered historical domains [2]. That result established a negative boundary: an anatomically meaningful and reproducible observation need not determine its own future.

The natural constructive response is to add memory. For a coarse history

$$H_h(x, t) = (O(F^t x), O(F^{t-1} x), \dots, O(F^{t-h} x)), \quad (1.1)$$

one asks whether equal histories have equal observed successors. The first registered experiment in this line evaluated orders $h = 0, 1, 2$ over a four-step exact orbit. Its frozen output appeared to give the desired transition:

$$h = 0, 1 : \text{failure}, \quad h = 2 : \text{historically labelled exact closed noninjective}. \quad (1.2)$$

That interpretation was wrong. The three orders in (1.2) were evaluated on different left boundaries. Order zero included $t = 0$, order one began at $t = 1$, and order two began at $t = 2$. A hostile common-support audit showed that, once all orders are restricted to the same source-time points $t \in \{2, 3\}$, they induce exactly the same successor-compatible partition. The past two observations add no distinction on that support.

The actual result is more structural than finite-memory restoration. The initial coarse equivalence relation contains classes that are incompatible with the dynamics. After the registered transient, those unsafe classes are absent from the reached-set slices. A different, disjoint family

of non-singleton classes remains, and those classes are successor-compatible. Thus an exact one-step quotient factorization becomes valid by restriction to a post-transient reached set rather than by enrichment with lag coordinates. The terminal image does not return to the verified coarse image, so this is not an endomap or an indefinitely iterable quotient system.

The noninjective part also has a simple model-relative explanation. An exact audit of all five persistent cohorts finds zero clipped network drive at every saved current time $t = 1, 2, 3$. On the common support, their updates are therefore pure leakage. The factorization remains exact and noninjective, but its identified windows do not demonstrate compensation between nonzero effective drives. This distinction separates a valid finite factorization from a stronger interpretation of the dynamics that realizes it.

A second registered computation asks what exact internal transition-layer features distinguish the two fiber fates. It does not recover one universal trigger at the defining $1 \rightarrow 2$ transition. It does find a stable exact regime contrast afterward: at $2 \rightarrow 3$ and $3 \rightarrow 4$, two predeclared clipping-sensitive predicates hold for every transient-obstruction pair and for no persistent-safe pair in every shared-stratum comparison cell. The result therefore separates an unresolved, potentially heterogeneous initial trigger from a sharply different post-separation internal signature.

The two fiber fates can be stated together:

The transient does not repair the original ambiguous fibers. It eliminates them. Exact noninjective coarse factorization is then carried by a disjoint family of persistent successor-consistent fibers.

Here elimination is fiber-relative. It means that the original unsafe equivalence classes do not survive as equivalence classes on the later registered slices because their members become coarsely distinguishable. It does not mean that microscopic states disappear or converge.

1.1 Contributions

This paper makes five bounded contributions.

1. It corrects the original history-order interpretation by comparing all orders on one support and proving an exact noninjective one-step factorization on the registered $t \in \{2, 3\}$ slices.
2. It distinguishes separation of the 51 obstruction sources from five disjoint persistent non-singleton cohorts.
3. It shows that the terminal image leaves the verified coarse image, so the factorization is not a coarse endomap.
4. It gives an exact 6,975-record pair census and a post-separation transition-layer contrast without assigning a cause to the initial split.
5. It audits the entire noninjective part of the common-support partition and shows that common linear leakage suffices to preserve its one-step equality, without a demonstrated nonzero-drive compensation mechanism.

2 Related Work and Novelty Boundary

Classical lumpability asks when a coarse partition of a Markov process retains a closed transition law [5]. Higher-order lumpability separates first-order closure from closure after retaining observation history [4]. The present system is deterministic and evaluated on a declared finite orbit-window domain, but the governing fiber condition is the same factorization principle: equal retained descriptions must have equal retained successors. This paper does not claim lumpability theory as new.

Exact dynamical coarse-graining can be obtained by constructing collective variables that preserve selected dynamical quantities [7]. Network quotient and graph-fibration approaches likewise require compatibility between the microscopic dynamics and the chosen network partition [3]. Here the coarse observation is not optimized to guarantee closure. It is the fixed superclass-mean map used in the source-addressed MaleCNS case study.

Connectome coarse-graining is often motivated by the need to obtain tractable mesoscopic descriptions from very large anatomical networks [6]. The MaleCNS data provide the anatomical and annotation surface for the present model [1, 8]. The registered dynamics remain a computational model on that source, not a claim that the coarse quotient is a physiological brain state.

Paper XVI supplies the immediate negative precursor. It separates structural liftability, signed operator semantics, observation visibility, and dynamic closure, and it gives exact witnesses showing that the registered coarse observation does not close on its initial historical domains [2]. The present paper does not modify those results. Its new object is the exact fiber geometry after restricting the same registered observation to later reached-set slices, together with an exact internal transition-layer census of the two resulting fiber fates.

The novelty boundary is consequently narrow:

This paper provides a source-addressed exact finite certificate that a lossy coarse observation, invalid as a one-step factor during an initial transient, supports an exact noninjective factorization after restriction to declared later reached-set slices, and it decomposes that transition into transient separation and disjoint persistent safe forgetting. It further gives an exact post-separation layer contrast between the two declared pair families without identifying a universal trigger for the initial split. A complete persistent-cohort cache audit establishes that all non-singleton fibers on the common support have zero clipped drive; their one-step equality is consequently explained by the common leak.

It does not provide a general eventual-lumpability theorem, a causal mechanism, or a new principle of clipping or scalar leakage. The exact coverage and its interpretive consequence are case-study results, not claims of active compensation, learning, or biological forgetting.

3 Finite Dynamic Descent

3.1 Fiber Criterion

Let $D \subseteq X$ be finite. The observation O descends the microscopic update F on D when there is a map

$$\overline{F} : O(D) \longrightarrow Z \quad \text{such that} \quad O \circ F = \overline{F} \circ O \quad \text{on } D. \quad (3.1)$$

Proposition 3.1 (finite one-step fiber criterion).

Such a map $\overline{F}$ exists if and only if

$$O(x) = O(y) \implies O(Fx) = O(Fy), \quad x, y \in D. \quad (3.2)$$

Proof. Necessity follows by applying $\bar{F}$ to the common observation. For sufficiency, define $\bar{F}(z) = O(Fx)$ using any $x \in D$ with $O(x) = z$; condition (3.2) makes the value independent of the representative.

The factorization is noninjective when at least one fiber of $O|_D$ has more than one microscopic representative. Factorization and reconstruction are therefore different requirements. Proposition 3.1 does not make $\bar{F}$ an endomap of $O(D)$; that stronger conclusion additionally requires $\bar{F}(O(D)) \subseteq O(D)$. $\square$

3.2 Orbit-Window Domains

For source $a \in D_0$, let

$$x_{a,0} = e_a, \quad x_{a,t+1} = F_Y(x_{a,t}), \quad z_{a,t} = O(x_{a,t}). \quad (3.3)$$

The order- h history and successor are

$$H_h(a, t) = (z_{a,t}, z_{a,t-1}, \dots, z_{a,t-h}), \quad S(a, t) = z_{a,t+1}. \quad (3.4)$$

For a source-time universe Ω , exact order- h successor factorization means

$$H_h(a, t) = H_h(b, s) \implies S(a, t) = S(b, s) \quad (3.5)$$

for all $(a, t), (b, s) \in \Omega$. Equality in (3.5) is canonical rational identity, not floating tolerance or hash equality.

3.3 Reached-Set Restriction

For time indices I , define the sampled reached set

$$R_I(D_0) = \{x_{a,t} : a \in D_0, t \in I\}. \quad (3.6)$$

The observation equivalence relation is

$$x \sim_O y \iff O(x) = O(y). \quad (3.7)$$

The quotient supports a one-step factorization on $R_I(D_0)$ precisely when $\sim_O$ is successor-compatible there. A quotient may therefore fail on the initial domain while succeeding after restriction to a later reached set. This is a source-domain statement; it does not imply that the successor lies in the same coarse image or that the quotient can be iterated indefinitely.

3.4 Common-Support Requirement

Suppose multiple history orders are compared. If order h is evaluated on

$$\Omega_h = \{(a, t) : a \in D_0, h \leq t < T\}, \quad (3.8)$$

then changing h changes both the representation and the source-time domain. An apparent order effect may therefore be a left-boundary effect. A valid memory comparison fixes $H_{\max}$ and evaluates every $h \leq H_{\max}$ on

$$\Omega_{H_{\max}, T}^{\text{com}} = \{(a, t) : a \in D_0, H_{\max} \leq t < T\}. \quad (3.9)$$

This requirement is central to the correction below.

4 Finite Model and Exact Domain

4.1 Signed Dynamics and Observation

The model uses the same node basis and signed normalized $Y_{\text{known}\pm}$ carrier as the dynamic line of Paper XVI. The stored carrier uses source rows and target columns; state propagation uses the registered incoming action $Y^\top x$. The exact deterministic update is

$$F_Y(x) = (1 - \alpha)x + \alpha \phi(\gamma Y^\top x), \quad \phi(u)_v = \min\{1, \max\{0, u_v\}\}, \quad (4.1)$$

with $\alpha = 1/5$, $\gamma = 1$, zero external input, and no stochastic noise. All registered coefficients and states are represented as reduced rationals.

The observation O is the superclass-mean vector. For the declared partition $V = \bigsqcup_i V_i$,

$$O_i(x) = \frac{1}{|V_i|} \sum_{v \in V_i} x_v. \quad (4.2)$$

The observation is an operational annotation aggregate, not an assumed biological state. The computational evidence starts from the retained exact orbit sidecar; microscopic orbit generation is not replayed by this package.

4.2 Reduced Domain and Selection

The complete eligible source universe contains 163,439 rows across 25 nonempty superclass strata. A declared resource model selected 711 sources. One source was first assigned to every nonempty stratum; the remaining seats used largest-remainder apportionment on remaining stratum capacity, and SHA-256 rank selected members within each stratum. The declared selection rule does not read trajectories, factorization, or candidate outcomes.

The reduced domain is a declared computational domain, not a random sample or a prevalence estimator. The retained immutable record does not independently establish that the scope record preceded the computed outcome.

4.3 Exact Horizon

The registered rectangle is

$$|D_0| = 711, \quad H = 2, \quad T = 4. \quad (4.3)$$

Each source contributes one exact orbit $z_{a,0}, \dots, z_{a,4}$, from which orders 0, 1, 2 are derived. Rational values are serialized canonically with positive reduced denominators. SHA-256 is an index and integrity mechanism; every potential equality is resolved by canonical payload comparison.

The resource gates were satisfied by the completed run. The scope, exact sidecar, and subsequent audits are bound in the paper-owned evidence package described in Appendix A.

5 Frozen Primary Result

The original order-specific classification is:

order h	source-time support	(N_h, K_h, M_h)	successor-fiber test
0	$0 \leq t \leq 3$	$(2844, 1819, 406)$	16 conflicts
1	$1 \leq t \leq 3$	$(2133, 1797, 51)$	6 conflicts
2	$2 \leq t \leq 3$	$(1422, 1228, 51)$	exact, noninjective

The original classification called order two minimal on these order-specific domains. The next section tests whether that interpretation survives a common source-time support.

The left boundary moves with h . In particular, all six order-one failure fibers occur at $t = 1$, a time omitted from the order-two domain. The frozen table therefore mixes two changes:

$$\text{history representation} \quad \text{and} \quad \text{source-time support.} \quad (5.1)$$

The next section separates them.

6 Common-Support Correction

Fix

$$\Omega^{\text{com}} = D_0 \times \{2, 3\}. \quad (6.1)$$

For each $h \in \{0, 1, 2\}$, build the exact fiber partition Π_h on this same set of 1,422 source-time points.

Computational Certificate 6.1 (common-support factorization). On Ω^{com} , orders $h = 0, 1, 2$ all satisfy the exact successor-fiber criterion. Each order has

$$N_h = 1422, \quad K_h = 1228, \quad M_h = 51, \quad (6.2)$$

ten non-singleton fibers, and zero failure fibers.

Computational Certificate 6.2 (partition identity). The three induced partitions are exactly identical:

$$\Pi_0 = \Pi_1 = \Pi_2. \quad (6.3)$$

The common partition digest is bound in the paper-owned exact audit. Equality in (6.3) is equality of source-time blocks after exact payload comparison, not equality of hash labels alone.

6.1 Consequences

First, current observation already determines the registered successor on the common support:

$$z_{a,t} = z_{b,s} \implies z_{a,t+1} = z_{b,s+1}, \quad (a, t), (b, s) \in \Omega^{\text{com}}. \quad (6.4)$$

Second, the lag coordinates add no partition refinement there. On this finite support, equal current observations already force equal one- and two-lag history tuples. The past coordinates are redundant for discrimination on the registered points.

Third, the factorization remains genuinely lossy. Since

$$K_0 = 1228 < 1422 = N_0, \quad (6.5)$$

factorization is not obtained by assigning a unique code to every source-time point. The finite-domain window-count reduction is

$$\frac{N_0 - K_0}{N_0} = \frac{194}{1422} \approx 0.1364. \quad (6.6)$$

This is a count of identified registered windows, not a bit-level compression ratio.

Certificates 6.1 and 6.2 therefore rule out the intended minimal-memory conclusion: the positive result is already present at order zero on the common support, while the original order effect is confounded with the moving left boundary.

6.2 Terminal-Image Boundary

Let $\mathcal{Z}_{2,3} = O(R_{\{2,3\}}(D_0))$. The factor map in Proposition 3.1 has source $\mathcal{Z}_{2,3}$ and target Z . To obtain an autonomous coarse system on the same image, one would additionally need the registered successors to remain in $\mathcal{Z}_{2,3}$.

Computational Certificate 6.3 (terminal-image boundary). The exact observation sets at $t = 2$, $t = 3$, and $t = 4$ each contain 614 values. The $t = 2$ and $t = 3$ sets are disjoint, so $|\mathcal{Z}_{2,3}| = 1228$. All $t = 2$ successors lie in $\mathcal{Z}_{2,3}$, giving one registered internal shift. All 614 $t = 3$ successors lie outside $\mathcal{Z}_{2,3}$.

Consequently, the certificate establishes a horizon-bounded one-step factorization with one internal shift. It does not establish an endomap $\mathcal{Z}_{2,3} \rightarrow \mathcal{Z}_{2,3}$ or sustained coarse iteration.

7 Transient-Fiber Decomposition

The common-support result identifies where factorization holds but not how the initial obstruction disappears. A second result-owned audit compares the $t = 1$ obstruction sources with the non-singleton fibers at $t = 2$ and $t = 3$.

7.1 Transient Separation

The six order-one failure fibers contain respectively

$$9, 33, 2, 2, 2, 3 \quad (7.1)$$

sources, for a total of 51. Within each failure fiber, the sources have equal registered one-lag histories at $t = 1$ but distinct successors.

Computational Certificate 7.1 (transient separation). Every source in the six $t = 1$ failure fibers has a globally singleton coarse observation at $t = 2$ among all 711 registered sources.

Thus none of the 51 obstruction sources enters a post-transient non-singleton cohort. The obstruction is not repaired by appending a lag on the common support. The offending classes cease to exist as coarse classes because their members become distinguishable under O .

7.2 Persistent Safe Forgetting

The noninjectivity at $t = 2$ has a different source. Exactly five non-singleton source cohorts occur, with sizes

$$51, 25, 19, 4, 3. \quad (7.2)$$

Computational Certificate 7.2 (persistent successor-compatible cohorts).

The five source membership sets in (7.2) are identical at $t = 2$ and $t = 3$. Each cohort has one exact coarse observation at each time, and its registered next observation is common to all members. The cohorts are disjoint from the 51 transient-obstruction sources.

Their redundancy on each time slice is

$$(51 - 1) + (25 - 1) + (19 - 1) + (4 - 1) + (3 - 1) = 97. \quad (7.3)$$

The two registered slices therefore account for all

$$2 \times 97 = 194 \quad (7.4)$$

redundant windows in the common-support partition. The ten non-singleton fibers are precisely two time slices of five persistent source cohorts. Here “safe forgetting” denotes only the retention of a successor-compatible, noninjective observation on those slices. It does not name a learned or actively corrective process; Section 7.4 identifies its effective-drive boundary in the registered model.

7.3 Exact Decomposition

The finite registered transition can now be expressed as

$$\begin{array}{c} \boxed{\begin{array}{c} 51 \text{ sources in unsafe } t = 1 \text{ fibers} \\ \downarrow \\ 51 \text{ globally singleton coarse observations at } t = 2 \end{array}} \end{array} \quad (7.5)$$

together with the disjoint branch

$$\boxed{\begin{array}{c} \text{five non-singleton cohorts at } t = 2 \\ \downarrow \\ \text{the same five successor-compatible cohorts at } t = 3. \end{array}} \quad (7.6)$$

Equations (7.5) and (7.6) distinguish transient separation from persistent safe forgetting.

7.4 Effective-Drive Boundary

A post-result audit reads all 306 saved caches of the five persistent cohorts: 102 sources at $t = 1, 2, 3$. It verifies their frozen inventory bindings and recomputes clipping from the exact raw-drive payloads, without regenerating $Y^\top x_t$ or microscopic orbits.

Computational Certificate 7.3 (persistent effective-drive boundary). Every saved raw drive is nonzero and coordinatewise nonpositive; every clipped drive is zero. Thus all 204 members of the ten non-singleton fibers on $t \in \{2, 3\}$ have zero clipped drive. These fibers account for all 194 redundant windows.

The update in (4.1) is therefore

$$F_Y(x_{a,t}) = \frac{4}{5}x_{a,t}, \quad C(Y^\top x_{a,t}) = 0. \quad (7.7)$$

For cohort members a, b , linearity of O gives

$$\begin{aligned} O(F_Y x_{a,t}) - O(F_Y x_{b,t}) &= \frac{4}{5}(O(x_{a,t}) - O(x_{b,t})) \\ &= 0. \end{aligned} \quad (7.8)$$

Exact finite fiber evolution

Registered 711-source domain | unsafe fibers separate; disjoint cohorts remain safely noninjective

A Transient separation

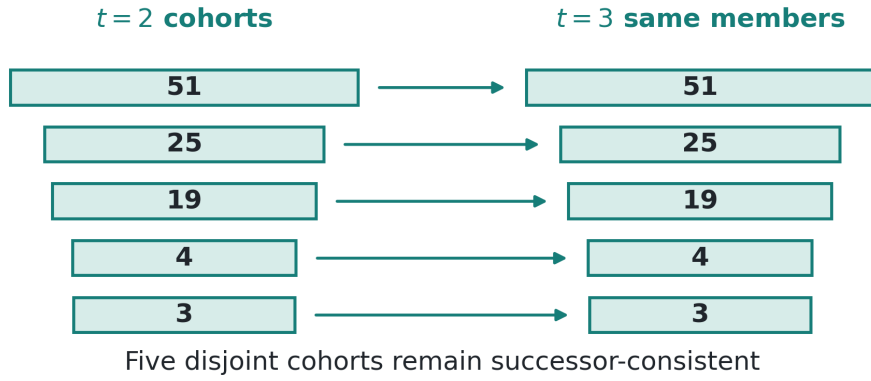
$t = 1$: **six unsafe coarse fibers**



$t = 2$: **51 globally singleton observations**



B Persistent safe forgetting



C Common-support factorization and terminal-image boundary

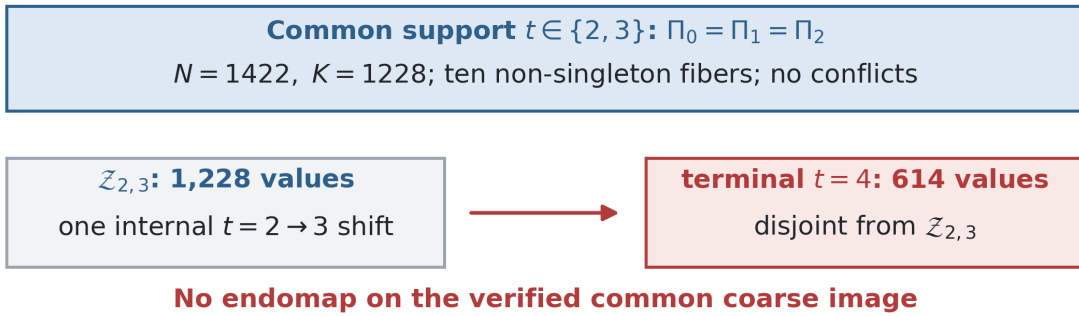


Figure 1. Exact finite fiber evolution. The six unsafe $t = 1$ fibers separate into singleton observations at $t = 2$, while five disjoint cohorts retain membership and successor compatibility from $t = 2$ to $t = 3$. The common-support partitions agree, but the terminal $t = 4$ image leaves the verified coarse image. Arrows denote only registered finite transitions.

Thus common leakage is a sufficient explanation of one-step equality on the entire noninjective part. No member there has a nonzero effective network drive, and no unequal nonzero clipped drives are being hidden by the observation map. The microscopic states themselves need not be equal: their differences are multiplied by $4/5$, not erased.

The $t = 1$ caches concern the same 102 sources but do not extend the common-support factorization to all $t = 1$ fibers. Nor is this a pure-leak classification of all 711 sources. Nonpositivity of the total drive does not establish nonpositivity of every individual source-block contribution or arbitrary block-subset robustness. The cache audit and (7.7)–(7.8) also do not make the finite coarse image forward invariant or certify an indefinitely iterable quotient.

8 Internal Transition-Layer Contrast

The fiber decomposition establishes two different coarse fates but does not identify their internal transition signatures. The transition-layer comparison uses the 51 transient-obstruction sources and 102 persistent-safe sources. External input and stochastic noise remain zero.

8.1 Exact Pair Census

Every unordered source pair is oriented canonically by source identifier, with $a < b$ and all signed differences defined as $\text{value}(a) - \text{value}(b)$. For one registered transition, write

$$\delta x = x^a - x^b, \quad \delta u = u^a - u^b, \quad u^a = Y^\top x^a, \quad u^b = Y^\top x^b. \quad (8.1)$$

The exact coarse residuals are

$$\begin{aligned} r_{\text{raw}} &= O(u^a) - O(u^b), \\ r_{\text{clip}} &= O(C(u^a)) - O(C(u^b)), \\ r_{\text{corr}} &= r_{\text{clip}} - r_{\text{raw}}. \end{aligned} \quad (8.2)$$

The census contains all 570 unordered pairs within the six frozen transient-obstruction fibers and all 1,755 unordered pairs within the five persistent-safe cohorts. Each pair is evaluated at $1 \rightarrow 2$, $2 \rightarrow 3$, and $3 \rightarrow 4$, giving

$$3(570 + 1755) = 6975 \quad (8.3)$$

pair-transition records, derived from 459 exact source-transition records for 153 distinct sources. The comparison design keeps transition time, transient cohort, persistent cohort, and shared initial anatomical stratum explicit, producing 15 nonpooled comparison cells.

The comparison uses seven fixed exact predicates. Write d_x and d_u for the exact microscopic-state and raw-drive L^1 differences, and $N_{\text{offdiag}}^{\text{fate}}$ for the number of coordinates with differing clipping-fate labels. In the registered bit order, its predicates are:

bit	predicate	exact condition
1	clipping creates a coarse residual	$r_{\text{raw}} = 0, r_{\text{clip}} \neq 0, r_{\text{corr}} = r_{\text{clip}}$
2	clipping eliminates a coarse raw residual	$r_{\text{raw}} \neq 0, r_{\text{clip}} = 0, r_{\text{corr}} = -r_{\text{raw}}$
3	clipping modifies a nonzero coarse residual	$r_{\text{raw}} \neq 0, r_{\text{clip}} \neq 0, r_{\text{corr}} \neq 0$
4	raw residual persists unchanged	$r_{\text{raw}} \neq 0, r_{\text{clip}} = r_{\text{raw}}, r_{\text{corr}} = 0$
5	raw-drive difference is coarsely annihilated	$d_u \neq 0, r_{\text{raw}} = 0$
6	microscopic difference has identical raw drive	$d_x \neq 0, d_u = 0$
7	fate disagreement with nonzero correction	$N_{\text{offdiag}}^{\text{fate}} \neq 0, r_{\text{corr}} \neq 0$

The exact conditions are evaluated on one oriented pair record; no coordinate is selected after observing a result. Bits 3 and 7 are decisive below:

$$\begin{aligned} P_{\text{mod}} : & \quad r_{\text{raw}} \neq 0, \quad r_{\text{clip}} \neq 0, \quad r_{\text{corr}} \neq 0, \\ P_{\text{fate}} : & \quad N_{\text{offdiag}}^{\text{fate}} \neq 0, \quad r_{\text{corr}} \neq 0, \end{aligned} \tag{8.4}$$

These are conjunctions of registered exact quantities. Neither a residual quadrant alone nor the identity relating r_{clip} to the observed successor can award the mechanism outcome.

8.2 Exact Finite Contrast

Computational Certificate 8.1 (complete registered contrast). The 15 cells, indexed by transition and stratum, have distinct exact normalized histograms of the seven predicates. Equality is checked by integer cross multiplication. No floating tolerance, prevalence threshold, majority rule, or hypothesis test is used.

Computational Certificate 8.2 (post-separation layer localization). At each of $2 \rightarrow 3$ and $3 \rightarrow 4$, both P_{mod} and P_{fate} hold for every transient-obstruction pair and for no persistent-safe pair in every shared-stratum comparison cell. These give four exact separators: two predicates across two transitions, always in the same direction.

The cellwise comparison does not cover pairs outside shared initial strata. An exact full-cohort tally gives the global counts:

transition	predicate	transient true / total	persistent true / total
$2 \rightarrow 3$	P_{mod}	570 / 570	0 / 1,755
$2 \rightarrow 3$	P_{fate}	570 / 570	0 / 1,755
$3 \rightarrow 4$	P_{mod}	570 / 570	0 / 1,755
$3 \rightarrow 4$	P_{fate}	570 / 570	0 / 1,755

The global tally also includes three transient and 180 persistent pairs outside the shared-stratum cells.

The result is a stable exact internal regime contrast. The former obstruction pairs continue to exhibit clipping modification of a nonzero coarse residual and clipping-fate disagreement with nonzero correction. The persistent-safe pairs exhibit neither registered predicate on those two transitions. This statement is relative to the declared finite pair universes and exact feature families; it is not a population or causal claim.

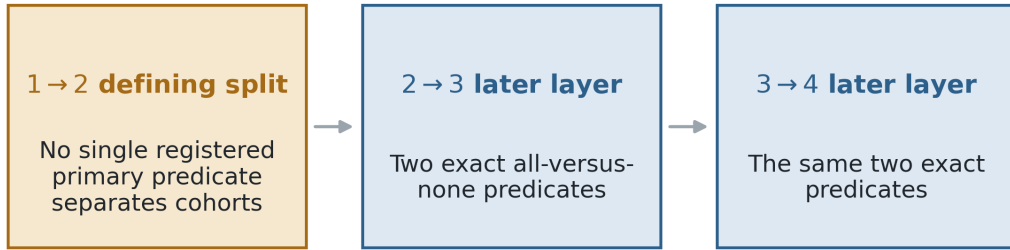
8.3 Initial-Trigger Boundary

No primary predicate is an all-versus-none separator at $1 \rightarrow 2$, although all five registered comparison-cell histograms differ there. The two later separators do not explain the original split. A heterogeneous initial trigger is possible, but this census does not establish one.

Stable internal transition-layer contrast after transient separation

TO = transient-obstruction pairs; PS = persistent-safe pairs. Exact finite-census comparison.

A Time-resolved interpretation boundary



Later signatures do not establish the cause of the initial split

B Exact registered separator matrix

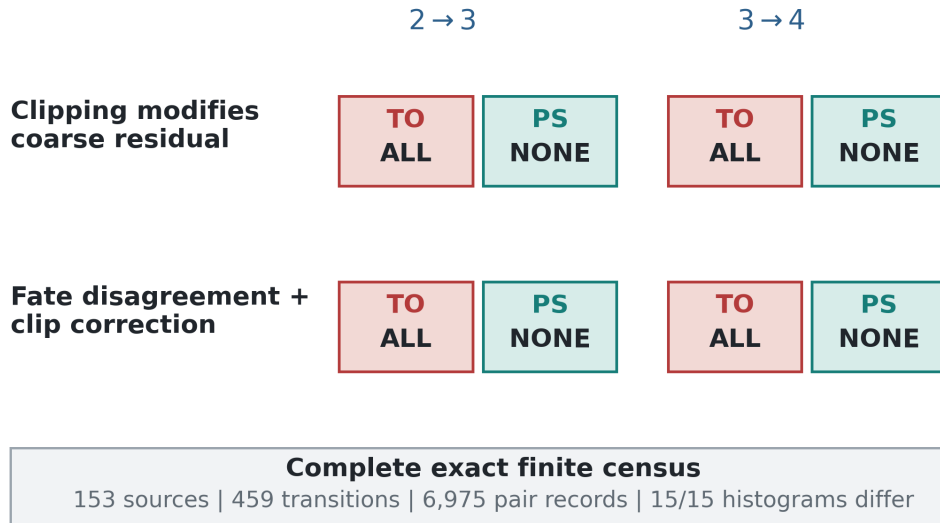


Figure 2. Exact registered transition-layer contrast. At $2 \rightarrow 3$ and $3 \rightarrow 4$, clipping modification of a nonzero coarse residual and fate disagreement with nonzero clipping correction hold for every transient-obstruction pair and no persistent-safe pair. The $1 \rightarrow 2$ histograms differ, but no single registered primary predicate separates the cohorts there; later signatures do not identify the initial trigger.

9 Claim Status and Boundary

result	status	scope
Proposition 3.1: fiber criterion	Proposition	A declared finite domain and observation
Certificates 6.1–6.2: common-support correction	Computational Certificate	711 sources, $t = 2, 3$, $h = 0, 1, 2$
Certificate 6.3: terminal-image boundary	Computational Certificate	Saved $t = 2, 3, 4$ observation sets
Certificates 7.1–7.2: two fiber fates	Computational Certificate	51 obstruction sources and five disjoint cohorts
Certificate 7.3: effective-drive boundary	Computational Certificate	102 sources; saved $t = 1, 2, 3$ caches
Certificates 8.1–8.2: transition-layer contrast	Computational Certificate	Frozen pair census and global tally
Cause of the defining initial split	Research Program	Not established by the retained contrast

The factorization, fiber decomposition, and transition-layer contrast are local exact computational certificates. They concern 711 selected sources and the registered horizon through $t = 4$; the 194-window reduction counts finite source-time identifications, not information bits. The later transition-layer separators identify a contrast between two declared pair families, not the cause of the initial $1 \rightarrow 2$ split.

The effective-drive audit covers all ten non-singleton fibers on the common support. Their one-step equality has the sufficient pure-leak explanation in (7.7)–(7.8); the paper does not demonstrate active compensation, learned equivalence correction, or rich nonzero-drive compression there. The $t = 1$ readout and later intervention studies do not enlarge the declared factorization domain.

No full-domain closure, coarse endomap, minimal memory order, eventual invariance, microscopic convergence, or physiological state claim follows. Local exact replay is not independent replication, and the retained record does not independently authenticate the scope record’s temporal precedence. Appendix A identifies the source-addressed claim and evidence closure.

10 Conclusion

The registered superclass mean fails one-step factorization during the initial transient but supports an exact noninjective factorization on the later registered slices. This is not a demonstrated memory effect: zero-, one-, and two-lag representations induce the same common-support partition. The six unsafe $t = 1$ classes disappear by coarse separation of their 51 members, whereas a disjoint family of five cohorts remains non-singleton and successor-compatible. The terminal $t = 4$ image leaves the verified common coarse image, so the certificate supplies one internal shift rather than a closed autonomous system.

The quotient thus becomes successor-compatible only after restriction to the registered reached set; neither history augmentation nor recovery of microscopic identity establishes this finite factorization. Its domain is part of the claim, not an incidental choice of sample.

The pair census adds a second exact result. All 15 comparison-cell histograms differ. At both later transitions, two clipping predicates hold for every obstruction pair and no persistent-safe pair. This establishes a stable internal contrast after separation. Neither predicate uniformly separates these families at $1 \rightarrow 2$, so the initial split remains unexplained. The remaining question is which internal routes first expose unsafe distinctions at the defining transition. This paper does not identify that initial cause.

The persistent side has a more definite interpretation. The exact drive audit covers the whole noninjective part of the common-support partition: every member has zero clipped network drive, so common leakage preserves one-step equality. The valid finite quotient therefore records reached-set restriction and drive-off persistence, not a demonstrated mechanism of active compensation or nonzero-drive dynamic compression.

Appendix A Computational Artifacts

The paper-owned evidence package is indexed at [RIME Lite, experiments/paper17](#). It binds the reduced-domain scope, exact orbit sidecar, common-support and fiber audits, transition-layer census, manuscript claim table, release manifest, and local validation receipts. The exact MTS-1 sidecar is deposited separately at [Zenodo DOI 10.5281/zenodo.23234143](#); this is a data-deposit DOI, not a DOI for the manuscript.

artifact	role	short path
A1: common-support audit	Exact partition and successor counts	<code>results/</code>
A2: fiber audit	Fixed cohort membership	<code>results/</code>
A3: effective-drive audit	Bound readout of 306 saved caches	<code>mechanism/</code>
A4: evidence table v2	Includes Certificate 7.3	Package root
A5: manifest v2	Current publication byte closure	Package root

The drive audit replays inventory-bound raw and clipped payloads without recomputing the operator action. Its optional full cache replay uses the deposited MTS-1 sidecar. The default publication gate checks compact bindings and coverage, not the remote sidecar bytes. Later intervention studies are separate research records, not additional claims of this paper.

All listed artifacts are available in the RIME repository. From its root, run the publication gate with the pinned Python 3.12.6 environment:

```
python experiments/paper17/validation/validate_release_v2.py
```

The default publication gate replays the retained finite-history classifications and checks compact MTS-1 and effective-drive bindings; it does not open the deposited source-cache or pair shards. With that sidecar materialized, separate read-only validators rederive all MTS-1 pair records, compare predicted successors with historical exact observations, and replay the 306 drive caches. These checks do not rerun microscopic orbit generation, independently replicate the study, or authenticate the temporal precedence of the declared scope record. The Zenodo binding checks deposit metadata and local reconstruction, not a remote full-byte download.

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